



CURRICULUM OVERVIEW

Data Scientist Path

Build 3 Data Projects

A complete, project-driven journey from messy raw data to real, working predictive models. Students don't just learn about data science — they clean real datasets, build compelling visualizations, and train machine learning models that make real decisions.

3

Courses

24

Lessons Total

3

Data Projects

13–18

Age Range

LEARNING PROGRESSION

13

Python for Data

Load, clean, and analyse real-world datasets using Python.



14

Data Visualisation

Turn raw data into compelling, interactive charts and dashboards.



15

Machine Learning Projects

Build predictive models that make real decisions from real data.

Learn by analysing, not by watching. Every lesson pairs a real concept with a real dataset — no toy examples, no invented numbers.

A portfolio, not just a certificate. Students finish with three real data projects they can present, explain, and keep building on.



COURSE 13 · LESSONS 1–8

Python for Data

Load, clean, and analyse real-world datasets using Python

Python 3

Pandas

NumPy

Jupyter Notebooks

OVERVIEW

Every data project starts with messy, real-world data. Your child learns to wrangle it into something useful — finding patterns, calculating stats, and drawing conclusions.

PORTFOLIO PROJECTS

- Sports performance analysis
- School grade analysis
- City population trends

WHY IT MATTERS

Real data is never clean. Learning to load, filter, and fix it in Pandas is the single most-used skill in data science — the foundation every chart, model, and insight in this path depends on.

LESSON BREAKDOWN

- 1 Why data science matters (real examples)
- 2 Python refresher and Jupyter setup
- 3 Loading data from CSV and Excel
- 4 Pandas: selecting, filtering, sorting
- 5 Cleaning data: nulls, duplicates, types
- 6 Aggregations: group by, pivot tables
- 7 Statistical summaries and correlation
- 8 Full analysis project with a real dataset



COURSE 14 · LESSONS 1–8

Data Visualisation

Turn raw data into compelling, interactive charts and dashboards

Python

Matplotlib

Seaborn

Plotly

Dash

OVERVIEW

Data without visuals is just numbers. Your child learns to create charts that tell stories — the kind that make people say "I never knew that" when they look at the data.

PORTFOLIO PROJECTS

- Climate change dashboard
- Market trends chart
- Sports comparison visualiser

WHY IT MATTERS

A brilliant analysis nobody can read is wasted. This course teaches students to choose the right chart, design it clearly, and build interactive dashboards — the way data scientists actually communicate results.

LESSON BREAKDOWN

- 1 The science of good data visualisation
- 2 Matplotlib: line, bar, scatter plots
- 3 Seaborn: statistical visualisations
- 4 Choosing the right chart for your data
- 5 Color, labels, and storytelling
- 6 Interactive charts with Plotly
- 7 Building a Dash dashboard
- 8 Final: your own interactive dashboard



COURSE 15 · LESSONS 1–8

Machine Learning Projects

Build predictive models that make real decisions from real data

Python

scikit-learn

XGBoost

Joblib

OVERVIEW

The pinnacle of data science: teaching a computer to predict outcomes. Your child builds real machine learning models — the same technology behind Netflix recommendations and fraud detection.

PORTFOLIO PROJECTS

- House price predictor
- Customer churn predictor
- Movie recommendation engine

WHY IT MATTERS

This is where cleaned data and clear charts become real predictions. Students learn the complete machine learning workflow — training, evaluating, tuning, and deploying a model — the same process used in industry.

LESSON BREAKDOWN

- 1 How machine learning works (no maths jargon)
- 2 Types of ML: regression vs classification
- 3 Train/test splits and model evaluation
- 4 Linear regression from scratch
- 5 Decision trees and random forests
- 6 Feature engineering and selection
- 7 Model tuning and improving accuracy
- 8 Deploying a prediction model as an API



The Complete Journey

Across 3 courses and 24 lessons, students progress from loading a messy spreadsheet to training a deployable prediction model — walking away with three real, working data projects for their portfolio.

COURSE 13

Python for Data

- Loading data from CSV & Excel
- Filtering, cleaning & fixing data
- Group-bys and pivot tables
- Statistical summaries & correlation

COURSE 14

Data Visualisation

- Matplotlib & Seaborn charts
- Choosing the right visualisation
- Interactive charts with Plotly
- Building a live Dash dashboard

COURSE 15

Machine Learning Projects

- Regression vs classification
- Decision trees & random forests
- Feature engineering & tuning
- Deploying a model as an API

WHAT STUDENTS BUILD

1

Python for Data Portfolio

Sports performance analysis ·
School grade analysis · City
population trends

2

Visualisation Portfolio

Climate change dashboard ·
Market trends chart · Sports
comparison visualiser

3

Machine Learning Portfolio

House price predictor · Customer
churn predictor · Movie
recommendation engine

BY THE END OF THE PATH, STUDENTS CAN



Load, clean, and analyse a real-world dataset using Python and Pandas.



Design clear, compelling charts and build an interactive data dashboard.



Train, evaluate, and deploy a real machine learning prediction model.

Ready to start building?

All lessons and projects are taught live by an expert Lumexa instructor.